

Factory Method and Strategy Pattern

Step-by-step runtime workflow with filename and exact line of code only.

Factory Method Pattern runtime workflow

Step	Role	File	Line	Code
1	Client	app/routers/upload.py	29	<code>validator = audio_validator_factory.create_validator(file.filename)</code>
2	Creator	app/services/audio_validation.py	47	<code>suffix = Path(filename or "").suffix.lower()</code>
3	ConcreteCreator	app/services/audio_validation.py	49	<code>return WavAudioValidator()</code>
4	Product	app/routers/upload.py	30	<code>await validator.validate(file)</code>
5	ConcreteProduct	app/services/audio_validation.py	23	<code>filename = Path(upload_file.filename or "").name.strip()</code>

Strategy Pattern runtime workflow

Step	Role	File	Line	Code
1	Context created	app/services/separation_service.py	138	<code>engine = self.engine_class(algorithm=algorithm, device_name="cpu")</code>
2	Context calls Strategy	app/ml/separation_engine.py	29	<code>return self.algorithm.separate(</code>
3	ConcreteStrategy	app/ml/neossnet_strategy.py	23	<code>if model_config_path is None:</code>